实验

一、数据集描述：

数据集的每一行对应一个用户的不同购买记录 transaction，由多个 item 构成。

例如：

1 2 4 8 9

4 9 10

上面的记录分别代表两个用户的购买信息，每个以空格切分的数字表示一个 item，也就是商品；每一行购买的商品形成一位用户的购买信息，也就是 transaction 数据。

注意：每个用户的 transaction 数据长度可能不同。

二、发布或查询的问题：

发布所有 item 的频数分布信息，即 item1，……， itemd 分别被多少人次购买。

例如：继续上面的例子，假设只有两条用户数据

需要发布 1，1，0，2，0，0，0，1，2，1

分别表示商品 1，2，3，4，5，6，7，8，9，10 被购买的次数。

三、实际应用场景：

①假设有**可信第三方**存在，设计发布方案，保护个人购买信息不被泄露。

提示：在差分隐私下利用拉普拉斯机制实现对最后频数估计数据的扰

动。（拉普拉斯机制参考3-12 页）

②假设**无可信第三方**存在，设计发布方案，保护个人购买信息在传输过程中不被泄露。

提示：在本地化差分隐私下利用 randomized response 机制或 UE 机制对每一个用户的数据进行扰动。（两种机制参考13-15页）

由于每个用户包含多个 item，并且长度不同，为了保护隐私，可随机采一个 item，调用扰动机制发布。

四、评估指标：

测量不同隐私预算 ε=（0.1，1，2，4）下，两种背景下的数据精度，精度用 MSE（平均平方误差）来衡量，定义为 $\frac{\sum_{i=1}^{d}\left(f_i - f_i^*\right)^2}{d}$，其中 f_i 和 f_i^* 分别表示精确频数与发布的带噪频数，d 表示总共的 item 数量。

五、实验报告要求：

①敏感度分析。

②方案设计的简要描述，1、2、3、4 步骤的形式。

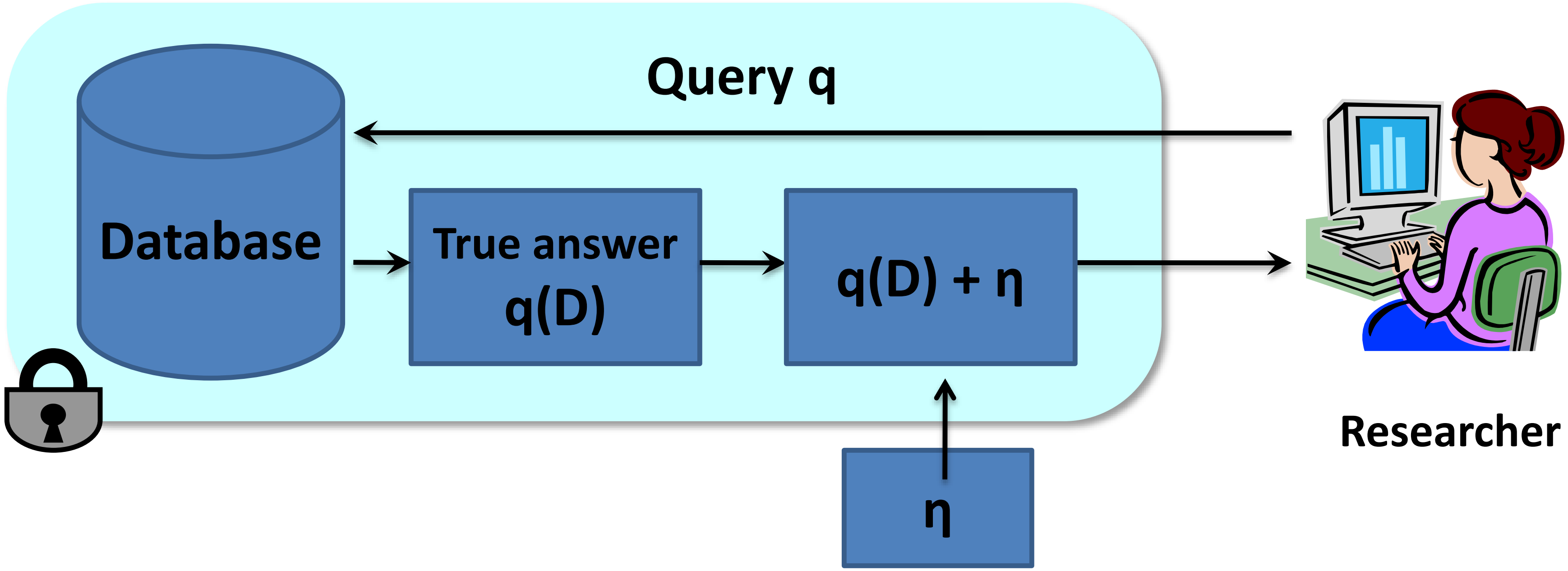
③两种场景下实验结果（MSE）对比，以曲线图或表的方式展现两种场景下在不同隐私预算 ε下的误差。

④代码粘贴到 word 中。

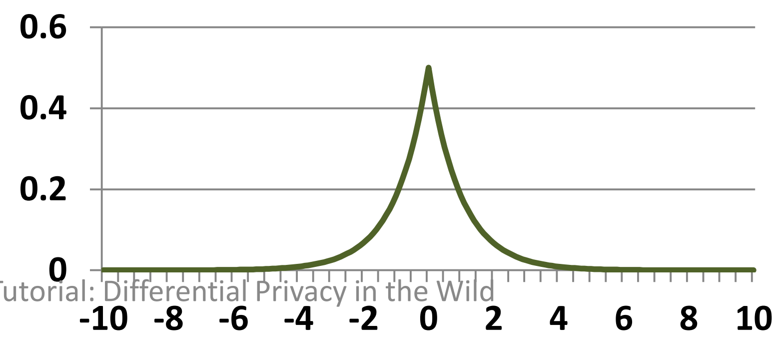
截止期：9.15 日前将实验报告发到邮箱 zhaoshuo@ouc.edu.cn，主题以及 word 的命名为“大数据隐私保护-学号-姓名”。逾期无成绩。

[DMNS 06]

Laplace Mechanism

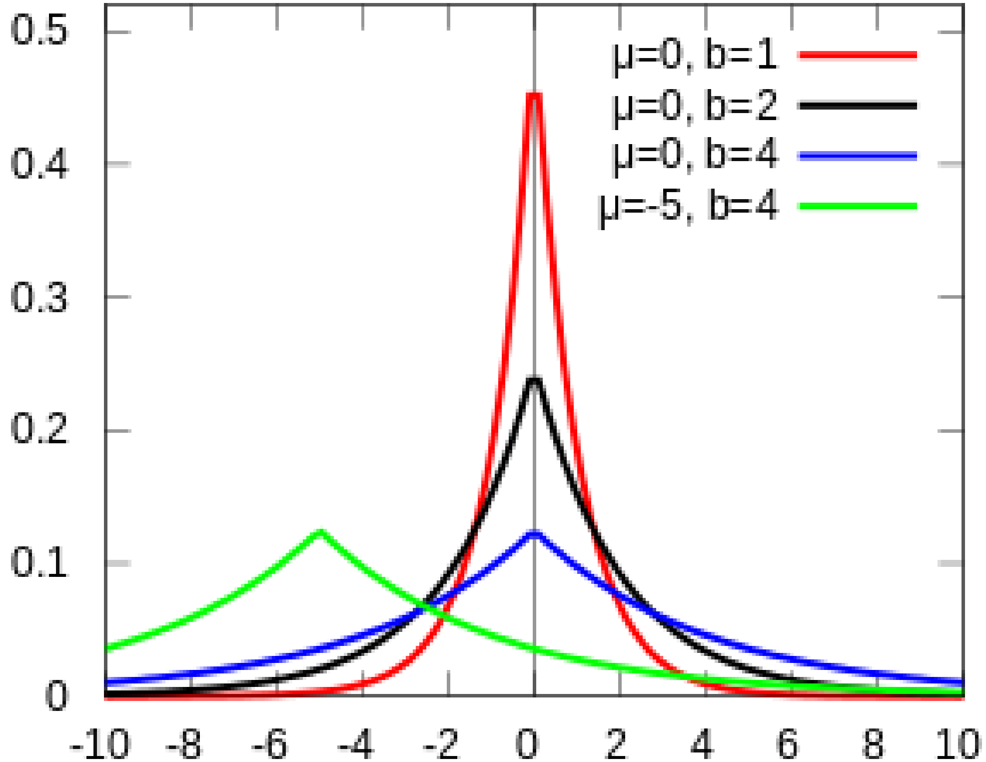


Laplace Distribution – Lap(S/ε)



Laplace Distribution

- PDF: $$f(x \mid \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b} \right)$$
- Denoted as Lap(b) when u=0
- Mean u
- Variance $2b^2$



How much noise for privacy?

[Dwork et al., TCC 2006]

Sensitivity: Consider a query q: *I* → R. S(q) is the smallest number s.t. for any neighboring tables D, D',

$$| q(D) - q(D') | \leq S(q)$$

Theorem: If **sensitivity** of the query is **S**, then the algorithm

A(D) = q(D) + Lap(S(q)/ε) guarantees ε-differential privacy

Privacy of Laplace Mechanism

- Consider neighboring databases D and D’
- Consider some output O

$$\frac{\Pr\left[A(D)=O\right]}{\Pr\left[A(D')=O\right]}=\frac{\Pr\left[q(D)+\eta=O\right]}{\Pr\left[q(D')+\eta=O\right]}$$

$$=\frac{e^{-|O-q(D)|/\lambda}}{e^{-|O-q(D')|/\lambda}}$$

$$\leq e^{\left|q(D)-q\left(D'\right)\right|/\lambda}\leq e^{S(q)/\lambda}=e^{\varepsilon}$$

Example: COUNT query

- Number of people having disease
- True answer = 3
- Sensitivity = ?

D

Disease (Y/N)
Y
Y
N
Y
N
N

Example: COUNT query

- Number of people having disease
- True answer = 3
- Sensitivity = 1

- Solution: $3 + \eta$,
  where η is drawn from Lap($1/\varepsilon$)
  - Mean = 0
  - Variance = $2/\varepsilon^2$

D

Disease (Y/N)
Y
Y
N
Y
N
N

Example: Sum (Average) query

- Sum of Age (suppose Age is in [a,b])
- Sensitivity = ?

Name	Age	HIV+
Frank	42	Y
Bob	31	Y
Mary	28	Y
Dave	43	N
…	…	…

Example: Sum (Average) query

- Sum of Age (suppose Age is in [a,b])
- Sensitivity = b

Name	Age	HIV+
Frank	42	Y
Bob	31	Y
Mary	28	Y
Dave	43	N
…	…	…

Utility of Laplace Mechanism

- Laplace mechanism works for **any function** that returns a real number
- Error: E(true answer – noisy answer)^2

$$= \mathrm{Var}(\ \mathrm{Lap}(S(q)/\varepsilon)\ )$$
$$= 2 * S(q)^2 / \varepsilon^2$$

- Error bound: very unlikely the result has an error greater than a factor (Roth book Theorem 3.8)

Outline

- Differential Privacy Definition
- Basic techniques
  - Laplace mechanism
  - Exponential mechanism
  - Random Response
- Composition theorems

Generalized Random Response (Direct Encoding)

- User:
  - Encode $x = v$ (suppose v from $D = \{1,2,\ldots,d\}$)
  - Toss a coin with bias p
  - If it is head, report the true value $y = x$
  - Otherwise, report any other value with probability $q = \frac{1-p}{d-1}$ (uniformly at random)
    - $p = \frac{e^{\varepsilon}}{e^{\varepsilon}+d-1}, q = \frac{1}{e^{\varepsilon}+d-1} \Rightarrow \frac{\Pr[P(v)=v]}{\Pr[P(v\prime)=v]} = \frac{p}{q} = e^{\varepsilon}$
- Aggregator:
  - Suppose n_v users possess value v, I_v is the number of reports on v.
  - $E[I_v] = n_v \cdot p + (n - n_v) \cdot q$
  - Unbiased Estimation: $c(v) = \frac{I_v - n \cdot q}{p-q}$

Intuitively, the higher p, the more accurate

However, when d is large, p becomes small (for the same ε)

ε	$p(d = 2)$	$p(d = 8)$	$p(d = 128)$	$p(d = 1024)$
0.1	0.52	0.13	0.016	0.001
1	0.73	0.27	0.027	0.002
2	0.88	0.51	0.057	0.007
4	0.98	0.88	0.307	0.05

To get rid of dependency on domain size, we move to the unary encoding protocols.

Unary Encoding (Basic RAPPOR)

- Encode the value v into a bit string $\boldsymbol{x} \coloneqq \vec{0}, \boldsymbol{x}[v] \coloneqq 1$
  - e.g., $D = \{1,2,3,4\}, v = 3,$ then $\boldsymbol{x}$ = [0,0,1,0]
- Perturb each bit, preserving it with probability p
  - $p_{1\to1} = p_{0\to0} = p = \frac{e^{\varepsilon/2}}{e^{\varepsilon/2}+1}$ $\quad p_{1\to0} = p_{0\to1} = q = \frac{1}{e^{\varepsilon/2}+1}$
  - $\Rightarrow \frac{\Pr[P(E(v))=\boldsymbol{x}]}{\Pr[P(E(v'))=\boldsymbol{x}]} \leq \frac{p_{1\to1}}{p_{0\to1}} \times \frac{p_{0\to0}}{p_{1\to0}} = e^{\varepsilon}$
    - Since $\boldsymbol{x}$ is unary encoding of v, $\boldsymbol{x}$ and $\boldsymbol{x}'$ differ in two locations
- Intuition:
  - By unary encoding, each location can only be 0 or 1, effectively reducing d in each location to 2. (But privacy budget is halved.)
  - When d is large, UE is better than DE.
- To estimate frequency of each value, do it for each bit.